

Emmanuel Seun SHITTU

Backend Engineer

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PROFILE

Backend Engineer with strong experience building fintech and payment-driven systems using Node.js, TypeScript, and modern backend frameworks. Proven track record designing scalable APIs, processing transactions, and integrating payment gateways (Paystack, PayPal, Monnify) with secure webhook architectures.

Experienced in event-driven systems, Redis-based job queues, and backend performance optimization (improving API response times by up to 75%). Skilled in PostgreSQL and MySQL database design, caching strategies, and deploying production systems using Docker and AWS with CI/CD pipelines.

Strong understanding of scalable backend architecture, modular system design, and microservices-oriented principles. Familiar with containerization (Docker) and orchestration concepts (Kubernetes), with hands-on experience in building production-ready backend systems.

Passionate about building secure, high-performance financial systems with clean architecture, reliability, and scalability in mind.

PROFESSIONAL EXPERIENCE

Divnerix Innovations *Backend Developer* 01/2025 – Present | Remote, Nigeria

- Designed and built scalable backend systems using Node.js and TypeScript
- Designed and implemented **transaction-safe systems** with idempotency and webhook reliability for financial operations
- Developed high-performance REST APIs for payments, donations, and event platforms
- Implemented event-driven architecture using Redis queues for background processing
- Integrated payment gateways (Paystack, PayPal, Crypto) with webhook handling
- Improved API performance by up to **40%** through caching and query optimization
- Deployed and managed applications on **AWS (EC2, S3)** with CI/CD pipelines
- Applied secure authentication using JWT and RBAC

ScholarGuide (EdTech Startup), *Backend Lead (Volunteer)* 02/2025 – 07/2025 | Remote, Nigeria

- Led backend architecture and API development for a multi-role EdTech platform
- Designed scalable systems supporting students, parents, teachers, and admins
- Built gamification features (points, badges, leaderboard) to improve engagement
- Implemented caching, rate limiting, and secure API practices
- Developed multi-tenant backend architecture for scalable user management
- Built JWT authentication and RBAC authorization system
- Documented APIs using Swagger for team collaboration

Ed3Hub – Web3-Enabled Learning Platform *Backend Developer (Contract)* 12/2024 – 03/2025 | Remote, Nigeria

- Designed and developed backend systems using Node.js and NoSQL databases
- Built scalable APIs for user management and learning features
- Implemented Redis caching to improve system performance
- Optimized database queries, improving performance by up to 75%
- Designed role-based data models and permission systems
- Contributed to backend architecture for platform scalability

GroovinX / SkyeStudio Project, *Backend Developer(Contract)* 10/2024 – 12/2025 | Remote

- Built backend systems for content automation and analytics
- Developed video processing pipelines using FFmpeg
- Designed APIs for content scheduling and automation workflows
- Optimized backend performance for reduced latency

Key Contributions

- Built **content scheduling and automation system** for media workflows
- Developed **video processing pipeline** using FFmpeg for media transformation
- Implemented **analytics backend system** for content performance tracking

In-Box Elite Nig LTD (Merchandising Company), *Full Stack Developer* 02/2023 – 12/2023 | Benin, Nigeria

- Developed full-stack features using React, Node.js, Express, and MySQL
- Built backend services for product, order, and inventory management
- Designed RESTful APIs for business operations
- Integrated frontend with backend systems
- Improved API performance by 30%

Key Contributions

- Built **end-to-end inventory and order management system**
- Automated internal business workflows improving efficiency by 40%
- Designed API layer for product lifecycle and order tracking
- Delivered full-stack integration for operational business systems

CareerEx, *Backend Development Intern* 06/2024 – 12/2024 | Remote

- Built RESTful APIs using Node.js and Express
- Integrated Paystack and PayPal payment systems
- Worked with SQL databases and structured backend logic
- Participated in Agile sprints and production deployments

Key Contributions

- Built **fintech-style payment integration APIs**
- Developed structured backend architecture for e-commerce systems
- Contributed to production-ready deployment workflows
- Strengthened API security and error handling systems

Full Stack Developer (Bootcamp Trainee and Intern)

- Built web applications using React and Node.js
- Developed REST APIs with Express and MongoDB
- Integrated frontend with backend systems
- Deployed applications using Vercel and Netlify

Key Contributions

- Built and deployed **7+ full-stack production-style applications**
- Developed real-time CRUD-based web applications
- Strengthened full-stack integration and API design skills

EDUCATION

Bachelor of Science, University of Benin [🔗](#)

10/2018 | Benin City, Edo State, Nigeria

Banking and Finance

National Diploma, Abraham Adesanya Polytechnic [🔗](#)

11/2007 – 06/2010 | Ijebu Igbo, Ogun State, Nigeria

Business Administration

SKILLS

Backend & Architecture

Node.js, Express.js, NestJS, TypeScript, REST APIs, Microservices, Event-Driven Systems

Security

JWT, RBAC, OAuth2, Rate Limiting, Secure API Design

Frontend

React.js, Next.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS

Testing & Tools

Jest, Supertest, Swagger (OpenAPI), Postman, Agile/Scrum

Databases & Caching

PostgreSQL, MySQL (strong), MongoDB, Redis (caching & queues), Query Optimization

DevOps & Cloud

Docker, Kubernetes (familiar with container orchestration concepts and deployment workflows), GitHub Actions (CI/CD), AWS (EC2, S3), Linux

CERTIFICATES

- Backend Development (NodeJs) – CareerEX
- Python and Statistics for Financial Analysis [🔗](#)
- Foundations: Data, Data, Everywhere [🔗](#)

PROJECTS

♦ Fintech Microservices Platform [🔗](#)

NestJS • TypeScript • Nx Monorepo • gRPC • RabbitMQ • Redis • BullMQ • PostgreSQL • Docker

GitHub: <https://github.com/shem4soul/Fintech-Microservices-Platform> [🔗](#)

- Designed and developed a distributed fintech platform using an Nx monorepo comprising Authentication, Wallet, Transaction, Integration, and Notification microservices.
- Implemented synchronous service-to-service communication using **gRPC** and asynchronous messaging with **RabbitMQ** for notification processing.
- Built a Transaction service that orchestrates wallet operations and coordinates requests across multiple microservices.
- Implemented a resilient retry mechanism using **BullMQ** and **Redis**, automatically retrying transient transaction failures before marking them as failed.
- Developed an Integration service that communicates with external APIs and caches exchange rate data in Redis to reduce latency and external API calls.
- Containerized the complete platform with Docker Compose, enabling consistent local development and deployment of all services.
- Applied distributed system patterns including service orchestration, event-driven messaging, caching, fault tolerance, and eventual consistency.

♦ Banking API (Wallets & Virtual Cards) [🔗](#)

- Implemented wallet funding, wallet-to-wallet transfers, withdrawals, and transaction history.
- Designed virtual card lifecycle operations including creation, blocking, unblocking, funding, and unloading.
- Applied CQRS to separate read and write operations.
- Built secure REST APIs with NestJS and PostgreSQL.
- Containerized the application with Docker Compose.

♦ Payment Parser & Executor API

Node.js, JavaScript, Layered Architecture, Middleware Systems

<https://github.com/shem4soul/payment-parser-executor-API> [🔗](#)

- Built a scalable backend system for parsing, validating, and executing structured payment instructions
- Designed layered backend architecture using endpoint, middleware, service, and repository separation patterns
- Implemented centralized error handling, structured logging, and reusable validation utilities
- Developed custom validation schema logic for request validation and business rule enforcement
- Built middleware pipelines for authentication, request logging, and request augmentation
- Structured backend modules by domain for maintainability and scalability
- Implemented clean service conventions and standardized API response handling
- Designed backend workflows focused on reliability, readability, and extensibility

♦ Payroll System (Monnify Integration)

Node.js, Express.js, TypeScript, PostgreSQL, Redis, Bull Queue

<https://github.com/shem4soul/Monnify-Payroll-System> [🔗](#)

- Built a production-style payroll processing system supporting bulk salary disbursement workflows
- Integrated Monnify bulk transfer APIs for automated payroll payments and transaction processing
- Implemented asynchronous payroll processing using Bull queues and Redis background workers
- Designed payroll lifecycle management including pending, processing, completed, and failed states
- Built webhook handlers for real-time payment status updates and transaction reconciliation
- Implemented transaction tracking, idempotency handling, and payroll statistics reconciliation
- Designed scalable PostgreSQL schemas for employees, payroll batches, and transaction records
- Developed secure REST APIs using TypeScript and Express.js

- Structured the system for reliability, scalability, and fault-tolerant payment processing

♦ **School Management System API**

Node.js, Express.js, MongoDB, JWT Authentication

 <https://github.com/shem4soul/school-management-system-API> 

- Built a comprehensive school management backend platform supporting students, teachers, administrators, exams, and academic operations
- Designed scalable REST APIs for managing academic terms, class levels, subjects, programs, exams, and results
- Implemented JWT-based authentication and role-based access control (RBAC) for Admin, Teacher, and Student workflows
- Developed secure exam and result management systems with controlled publishing workflows
- Structured backend modules using modular MVC architecture for maintainability and scalability
- Implemented secure CRUD operations, pagination, filtering, and centralized error handling
- Designed MongoDB schemas to support complex academic relationships and operational workflows
- Developed admin-only operations and protected routes for sensitive academic management tasks